

**ASSESSING THE EFFICACY OF SECTOR PORTFOLIO HRP: A
COMPARATIVE STUDY OF PERFORMANCE AND RISK METRICS
AGAINST THE MSCI ACWI**

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First version: November 3, 2023

Current version: November 3, 2023

Abstract

The present study examines the investing strategy we have termed as “Sector Portfolio HRP” and compares its performance measures with those of the MSCI All Country World Index¹ throughout the period of 1996 to 2022. The proposed approach encompasses a diversified portfolio consisting of 11 exchange-traded funds (ETFs) that are unique to various sectors. This strategy utilises the Hierarchical Risk Parity (HRP) method to adjust the weights of the ETFs on an annual basis.

In terms of performance, the HRP-centric method had more favourable outcomes compared to the MSCI ACWI. It achieved an annualised return of 7.56%, surpassing the latter’s return of 6.52%. Additionally, it accumulated a return of 614.88%, while the MSCI ACWI accumulated a return of 450.25%. The advantages of the strategy were further highlighted by a risk evaluation, which demonstrated reduced annualised volatility and a smaller maximum drawdown compared to the benchmark index.

In order to enhance the level of analysis, this study examined the linear correlations that exist among the utilised ETFs, evaluated their performance across smaller time periods, and utilised a range of risk indicators such as VaR, CVaR, and the Sortino Ratio. The HRP framework regularly demonstrated favourable outcomes across all parameters, particularly within the 5 and 10-year sub-periods.

It is important to highlight that the results obtained are based on specific market conditions and a limited historical dataset. The forward-looking applicability of these conclusions is intrinsically constrained, particularly due to the unpredictable nature of financial market dynamics.

The “Sector Portfolio HRP” strategy presents itself as an intriguing area for further academic exploration. Our initial findings highlight its inherent potential, but a more comprehensive analysis

¹ From now on, “MSCI ACWI”.

is required to fully understand its underlying dynamics and identify the optimal conditions for its implementation.

The research findings offer valuable and encouraging perspectives, which necessitate a cautious and thoughtful approach when implementing them in practical settings. We strongly support the need for further research efforts to develop a more comprehensive understanding of this methodology. This will enable the creation of investment strategies that are both resilient and sustainable over long periods of time.

Introduction and literature review

The field of financial investments has had significant advancements in recent decades with the emergence of diverse theories and models. The portfolio theory developed by Harry Markowitz in the early 1950s laid the foundation for important concepts such as asset allocation and diversification². It has been proposed that investors have the potential to improve their balance between return and risk by making savvy decisions in securities.

The concept at the core of this theory was further developed by William Sharpe through the introduction of the Capital Asset Pricing Model (CAPM)³. The aforementioned model incorporated the idea of 'beta', which serves as a measure of a stock's systematic risk, and therefore enhanced the principles pertaining to expected return and diversification.

Subsequently, Eugene Fama introduced the efficient market theory⁴, positing that asset prices intrinsically incorporate all available information. The theory presented in this study, which introduced the concepts of weak, semi-strong, and strong market efficiency, posed a significant challenge to the prevailing belief that it is possible to consistently outperform the market by selecting individual securities.

The following advent of the Arbitrage Pricing Theory (APT) has presented a flexible framework for the valuation of assets, enabling the consideration of different risk factors⁵.

The idea was initially associated with macroeconomic factors and later served as the foundation for factor analysis in the 1990s, led by Fama and French⁶. The "Three Factor Model" proposed

² Markowitz, 1952.

³ Sharpe, 1964.

⁴ Fama, 1969.

⁵ Ross, 1976.

⁶ Fama, French, 1992.

by the authors emphasised the significance of size, value, and market risk as key factors influencing the returns of securities.

After a few years, the three-factor model expanded to four⁷ and then to five factors⁸. By the year 2019, the scope of exploration had expanded to encompass more than 400 variables, leading to the popularisation of the phrase “factor zoo”⁹. The wide range of factors discussed has given rise to arguments questioning their true explanatory capacity: these factors might not be explaining anything and may be, instead, the result of many spurious analyses.

In recent years, there have been significant advancements in investment methods, with a focus on using quantitative methodologies, machine learning algorithms, and advanced statistical techniques. Significantly, the integration of artificial intelligence in the field of finance has represented a notable advancement, namely in the domain of portfolio construction.

We have made the decision to embrace an approach that aligns with Markowitz’s core thinking, rather than further exploring the complex network of models and methodologies.

The present study utilises the Hierarchical Risk Parity (HRP) hypothesis proposed by Marcos López de Prado¹⁰, which has attracted attention because of its ability to potentially overcome the limitations of Markowitz’s model.

Instead of identifying new factors, our approach taps into the inherent structure of financial markets, emphasising sectors and dynamically adjusting their weights.

⁷ Carhart, 1997.

⁸ Fama, French, 2015.

⁹ Harvey, Liu, 2019.

¹⁰ De Prado, 2018, pp. 221-238

The primary emphasis lies on sectors rather than factors, which represents a nuanced yet important differentiation in our analytical methodology.

The impact of the HRP model on sectorial portfolios remains a relatively unexplored domain in academic research. The primary objective of this study is to shed light on the aforementioned area by comparing the “Sector Portfolio HRP” with the performance of the MSCI ACWI index.

The aims of this study are threefold: to evaluate the effectiveness of the strategy in terms of its yield, to examine the potential risks associated with its implementation, and to provide conclusions that can inform future research efforts. After this introductory section, we will proceed to examine the current body of literature, present our methodological framework, and emphasise the insights derived from empirical data. The paper will finish by providing a summary of the findings, proposing potential areas for future research, and considering the practical implications.

Methodology

The analysis focuses on the HRP model and the market sectors that comprise the MSCI ACWI index.

Financial indices are tools for measuring market performance since they allow for the aggregation of large amounts of data using mathematical concepts. The first index to appear in 1884 was called the Dow Jones Railroad Average index, after its authors, Edward D. Jones and Charles H. Dow¹¹.

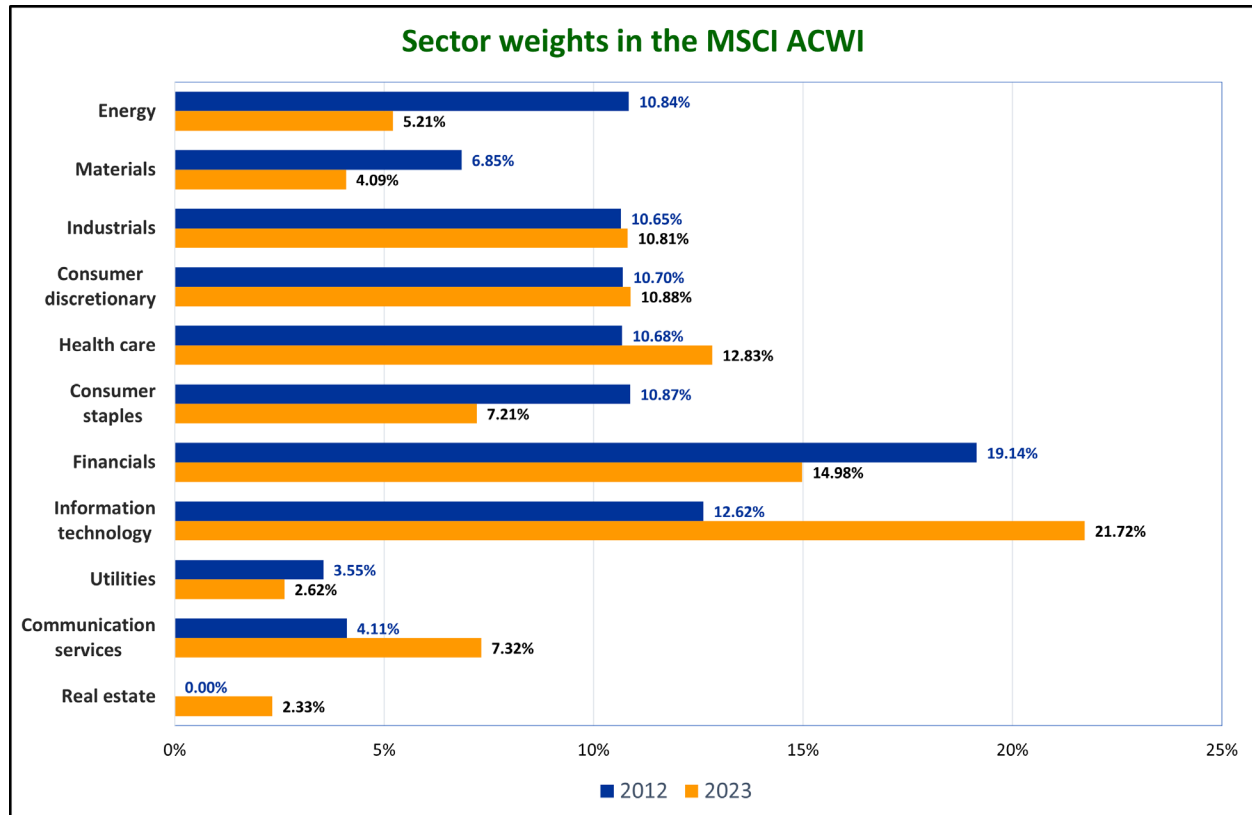
Today, investors have thousands of indices to choose from for measuring and monitoring the performance of financial markets and asset classes. Creating and managing these indices requires establishing the target market for replication, picking stocks within that market, and determining the sort of weight allocation for each security.

Index management entails rebalancing weights and reassembling the index's basket of assets. Rebalancing is essential to ensure that the weight of each security corresponds to the index's weighting technique. The process of revising stocks to reflect changes in the reference market or selection committee evaluations, on the other hand, is known as reconstitution.

Sectoral indexes represent and monitor consumer goods, energy, finance, health, technology, and so on. To classify companies, the sectoral indices that constitute the MSCI ACWI Investable Market Index (IMI) use the Global Industry Classification Standard (GICS®).

¹¹ It first appeared in the "Customer's Afternoon Letter", which would become "The Wall Street Journal" in 1889.

The graphic below compares the weight distribution of key sectors at the end of September 2023 to those recorded at the end of September 2012, eleven years earlier:



In our backtesting, which will be discussed later, we used 11 ETFs, one for each of the MSCI ACWI's 11 sectors:

1. EPRA (ISIN: LU1437018838).
 - Sector: Real Estate.
 - Tracked index: FTSE EPRA/NAREIT Developed.
2. XDW0 (ISIN: IE00BM67HM91).
 - Sector: Energy.
 - Tracked index: MSCI World Energy.

3. XDWC (ISIN: IE00BM67HP23).
 - Sector: Consumer Discretionary.
 - Tracked index: MSCI World Consumer Discretionary.
4. XDWF (ISIN: IE00BM67HL84).
 - Sector: Financials.
 - Tracked index: MSCI World Financials.
5. XDWH (ISIN: IE00BM67HK77).
 - Sector: Health Care.
 - Tracked index: MSCI World Health Care.
6. XDWI (ISIN: IE00BM67HV82).
 - Sector: Industrials.
 - Tracked index: MSCI World Industrials.
7. XDWM (ISIN: IE00BM67HS53).
 - Sector: Materials.
 - Tracked index: MSCI World Materials.
8. XDWS (ISIN: IE00BM67HN09).
 - Sector: Consumer Staples.
 - Tracked index: MSCI World Consumer Staples.
9. XDWT (ISIN: IE00BM67HT60).
 - Sector: Information Technology.
 - Tracked index: MSCI World Information Technology 20/35 Custom.
10. XDWU (ISIN: IE00BM67HQ30).
 - Sector: Utilities.

- Tracked index: MSCI World Utilities.

11. XWTS (ISIN: IE00BM67HR47).

- Sector: Communication Services.
- Tracked index: MSCI World Communication Services 20/35 Custom.

The historical series of the indexes mirrored by those ETFs were used in the periods preceding the inception date of each ETF.

The concept of optimising the weights of financial instruments in a portfolio to improve the expected return for a given risk or, conversely, to reduce the risk for a given expected return dates back to Harry Markowitz's work in 1952. In this context, we will utilise the MSCI ACWI index to analyse the effectiveness of our investing strategy.

Markowitz had a great insight: he was the first to recognise that different levels of risk may be associated with different optimal portfolios, which the investor chose based on their risk tolerance. His research exposed the financial world to a novel concept known as the “efficient frontier.”

The new portfolio construction process shows that the “optimal” portfolio is rarely the one that maximises the investor's expected return. Because of the covariance properties of the asset returns, the optimal solution is usually represented by a diversified portfolio.

Despite Markowitz's brilliant insight, the mathematical optimisation approach poses significant practical obstacles. The primary source of concern is from the fact that slight differences in projected returns can result in radically different portfolio compositions. Because future return projections are based on complex statistical procedures, differences between expected and actual returns are unavoidable. As a result, the so-called optimal portfolios created from these forecasts may not hold up in practice.

This problem can be traced back to Markowitz's procedure's use of quadratic optimisation. At some point, it is necessary to invert the return covariance matrix, which must be positively defined. This inversion is the source of estimate errors, especially when the conditional number of the covariance matrix of returns is large¹².

A covariance matrix's conditional number is the absolute value of the ratio between the matrix's highest and lowest eigenvalues. It rises when the portfolio's holdings become more correlated. The inverse covariance matrix becomes more unstable as the number of stocks in a portfolio increases and their correlation increases. Minor changes in input values can result in vastly different inverse matrices.

Marcos López de Prado termed this conundrum the "Markowitz's Curse"¹³: the more securities are correlated, the greater the need for diversification, yet the higher the likelihood of yielding an unstable outcome. Markowitz's Curse manifests itself by completely or partially offsetting the benefits of diversification as a result of estimating mistakes reflected in the return covariance matrix.

The HRP (Hierarchical Risk Parity) model seeks to overcome this issue by introducing the concept of hierarchy, which is completely missing from Markowitz's approach. The optimisation is divided into several hierarchical stages, with weights dispersed from the top down. These hierarchical frameworks are expected to produce more reliable and intuitive results. The HRP model makes use of information in the covariance matrix without requiring it to be positively defined or invertible.

¹² De Prado, 2018, p. 222.

¹³ De Prado, 2018, p. 222.

The optimisation algorithm progresses through three stages: tree clustering, quasi-diagonalisation, and recursive bisection¹⁴:

- Tree clustering: this introduces the concept of hierarchy.
- Quasi-diagonalisation rearranges the rows and columns of the covariance matrix so that higher values are closer to the diagonal.
- Recursive bisection: the algorithm determines the final weights of each asset using an inverse variance allocation.

During our R backtest, we allocated an initial subset of data to establish the input values for the HRP model. This resulted in the weights for each ETF in the in-sample phase. The performance of the portfolios formed with the prior weights in the out-of-sample phase was then evaluated.

We were able to quantify the results that would have been attained if the investment had been made using the Sector Portfolio HRP strategy by cyclically repeating this procedure, thus employing a multi-period portfolio management technique.

Let's look at the backtesting method to see how the analysis works. The full time range of the backtest is divided into 12-month sub-periods, each comparable to a calendar year. For example, in the sub-period between January and December 2018, we utilise the historical return series of the 11 sectorial ETFs for the year 2018 as input data for the HRP model at the beginning of January 2019. The HRP model will provide the ideal weights for each portfolio after processing the 2018 data.

¹⁴ De Prado, 2018, p. 224-231.

In essence, we estimated the returns and covariances based on the prior year's returns. We did not use any specific statistical techniques to ensure that the model was as simple as possible.

The performance of the portfolio, weighted as proposed by the HRP model, will be evaluated between January 2019 and December 2019. In January 2020, we repeat the previous year's operation, except this time we use the historical return series from 2019. The ideal weights for each portfolio will then be generated using the HRP model. We will perform these operations on an annual basis.

In summary, each year serves as the input data for the HRP model, which generates weights for specific ETFs. The portfolio performance analysis using these weights will be based on the succeeding period's returns.

This is a cyclical approach that allows us to generate a whole historical series of portfolio returns constructed using the methods mentioned above, encompassing the entire time period specified. The only exception is the first 12 months (1995), which are only used as input data for the HRP model during the initial backtesting phase.

Annual rebalancing in an investment strategy often seeks to control volatility. Its purpose is to keep the risk of the portfolio within the limitations defined and tolerated by the investor.

The argument about rebalancing frequency continues to this day: is it better to rebalance at periodic intervals or based on percentage thresholds of weight divergence from the baseline values?

What is the best rebalancing frequency in the first case? What is the most efficient threshold in the second? Because conclusive answers to these concerns have never been provided, we have decided to rebalance the portfolio on an annual basis. This option appears to achieve an ideal

compromise between maximising efficiency (frequent rebalancing) and limiting costs (less frequent rebalancing).

What distinguishes our rebalancing from traditional static rebalancing is that on each occasion, we initiated a new optimisation of the ETF weights using the previously assessed HRP model. As a result, it is a dynamic rebalance.

In essence, the HRP model is used to optimise the weight of financial instruments, and each new optimisation is likely to provide a different set of weights than the previous one. This is why we call the process dynamic rebalancing.

The analysis we conduct is based solely on portfolios generated using the information available at the time of their creation. This principle is critical: it is simple to make the mistake of using information that would not have been accessible at the start of the period under consideration while backtesting.

A financial model used for investments has the primary purpose of identifying specific variables that can provide important insights to improve the performance of the investment itself: either to increase returns, reduce risk, or achieve an ideal balance between the two.

To ensure predictive capability, the relationship between the variables considered in the model and the outputs generated, which will guide the investor's decisions, must be causal in nature.

Causality is defined as a "relationship of cause and effect"¹⁵ or a form of relationship that relates a certain action to a specific outcome. It is the type of relationship that allows us to predict a specific outcome as a result of a specific action we have taken.

¹⁵ Dictionary.com

In statistical terms, the correlation between two variables may indicate a causal relationship. Such an assumption, however, is far from guaranteed. Indeed, the correlation could be caused by random factors or an unobserved third variable, giving the association the appearance of being stronger (or weaker) than it is. To put it another way, correlation does not indicate causality.

For accurate data analysis, it's critical to distinguish between what can be considered evidence of causality and what cannot. In the real world, determining cause-and-effect relationships is almost never absolute. However, experimental, statistical, and research design techniques can give evidence to support causality. Randomisation, controlled experimentation, and the use of multivariate prediction models are among examples.

Randomisation is a method predominantly used in controlled clinical studies with the aim of minimising bias when testing a new treatment. Participants are randomly assigned to one of two groups: one that receives the experimental treatment and one or more control groups that get either a regular treatment or a placebo. This strategy attempts to counteract the selection effect by forming groups that are comparable to one another. The use of randomisation permits the treatment's effect to be isolated, eliminating distortion produced by other variables that may influence the outcome. In this way, it is possible to determine more precisely whether the observed effects may be attributable to the experimental treatment or not, while all other factors stay constant or are distributed equally among the groups.

A controlled experiment is an experimental methodology that divides participants into two groups: the experimental group, which receives the therapy under study, and the control group, which receives an alternative treatment or no treatment at all. The goal of good experimental design is to make the two groups as homogeneous as feasible, so that any differences in outcomes may be attributed to the treatment rather than other variables.

Finally, a multivariate predictive model is a mathematical model that predicts the value of a dependent variable using numerous independent factors. This statistical technique, also known as multiple regression, is frequently used to investigate the associations between distinct factors and to generate future predictions about the behaviour of a dependent variable based on the independent variables. This model allows for more intricate analysis and can provide a more complete picture of the phenomenon under consideration.

In economics and finance, using controlled experiments or randomisation techniques is frequently impractical or ethically objectionable. For example, it would be illegal to influence economic policy in an experimental environment or to withhold education to a control group in order to assess the impact of education on poverty.

Given this complexity, statistical techniques that analyse previous data are routinely utilised to study the underlying causes of numerous economic and financial phenomena. Multiple regression is one of the most commonly used techniques in this area. However, even this can lead to wrong or partially incorrect results.

From this standpoint, we believe there is a need to return to optimisation models that, at least mathematically, should work effectively. Starting with Markowitz's intuition, one can go into improving the performance of the MSCI ACWI index using Marco López de Prado's more recent Hierarchical Risk Parity (HRP) model.

Causality in this scenario can be attributed to the empirically validated relationship between expected return, variance, and expected covariances of the sectoral ETFs involved in the optimisation and portfolio performance: if estimates of these quantities are sufficiently accurate, the HRP optimisation should suggest an asset allocation that, when dynamically rebalanced, could outperform the index.

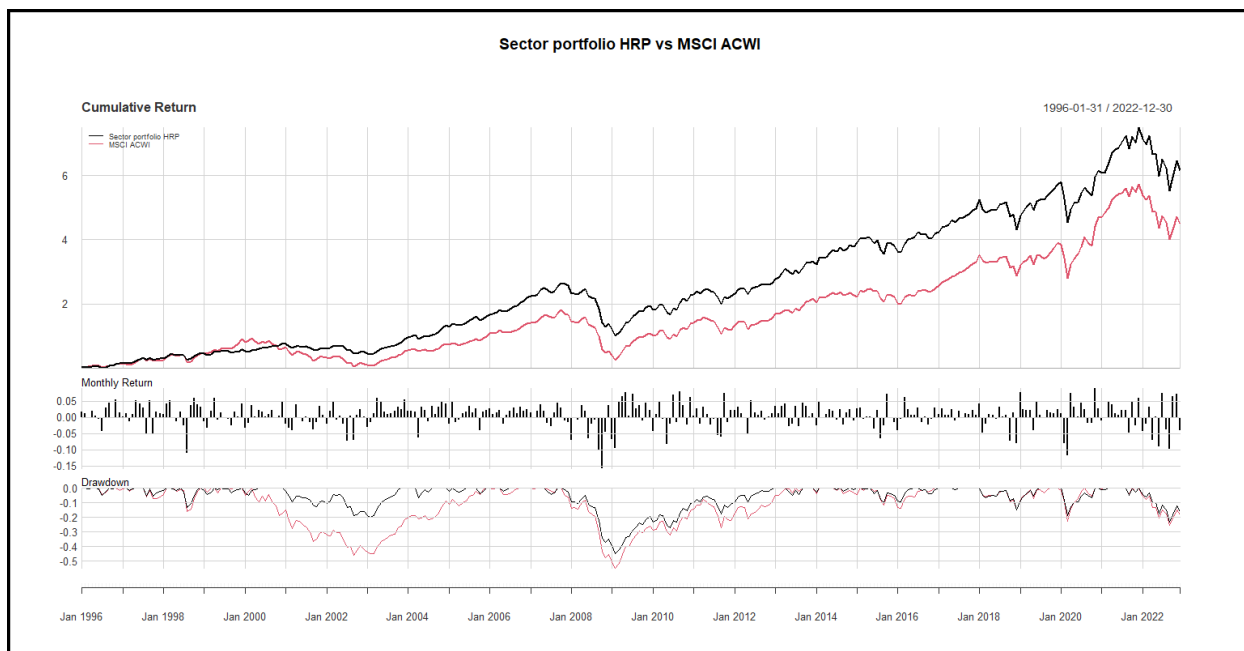
Results and discussion

The Sector Portfolio HRP strategy's performance, compared with the one of the MSCI ACWI index, is the following:

Performance 1996-2022

	Sector Portfolio HRP	MSCI ACWI
Annualized Return	7.56%	6.52%
Annualized Std Dev	12.61%	15.91%
Annualized Sharpe	0.5993	0.4096
Cumulative Return	614.88%	450.25%
Worst Drawdown	45.04%	55.11%
Average Drawdown	6.28%	8.41%
Average Length	7.41	9.10
Average Recovery	4.21	5.27
Ulcer Index	0.1054	0.1833
VaR	-6.04%	-7.64%
CVaR	-9.15%	-11.56%
Sortino Ratio	0.9237	0.6827

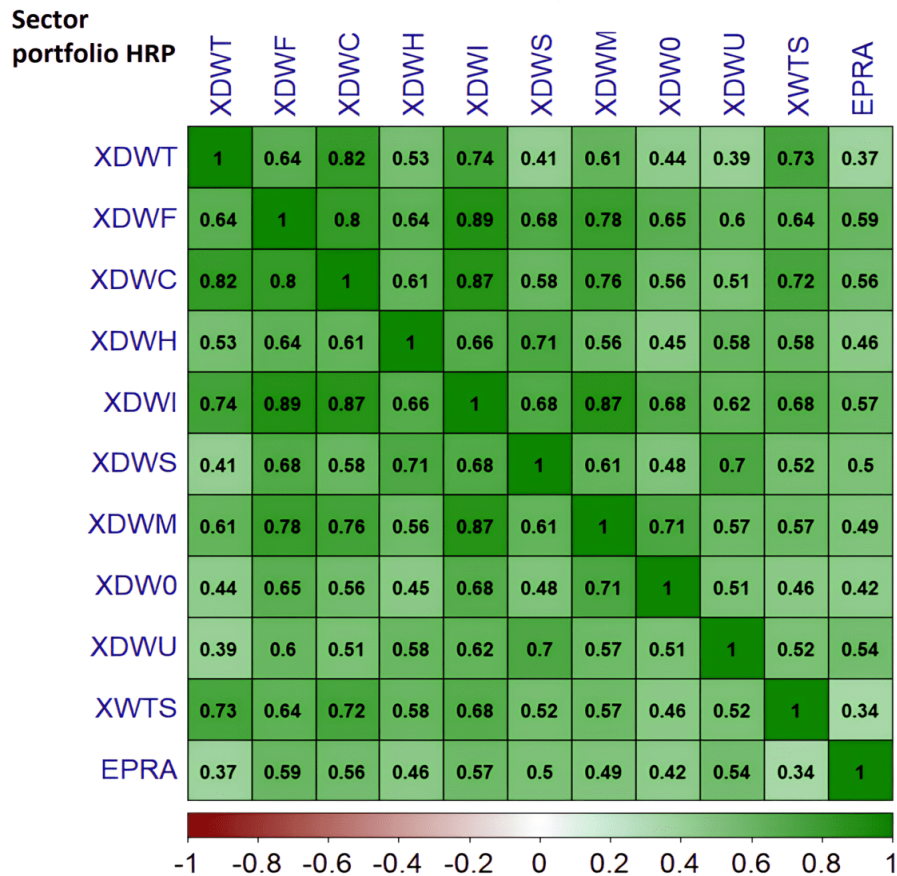
The chart below compares the Sector Portfolio HRP equity line to the MSCI ACWI equity line during the same time period. The good performance indicated in the preceding table is highlighted even more clearly in the equity line chart:



The black line represents the Sector Portfolio HRP equity, whereas the red line reflects the MSCI ACWI equity.

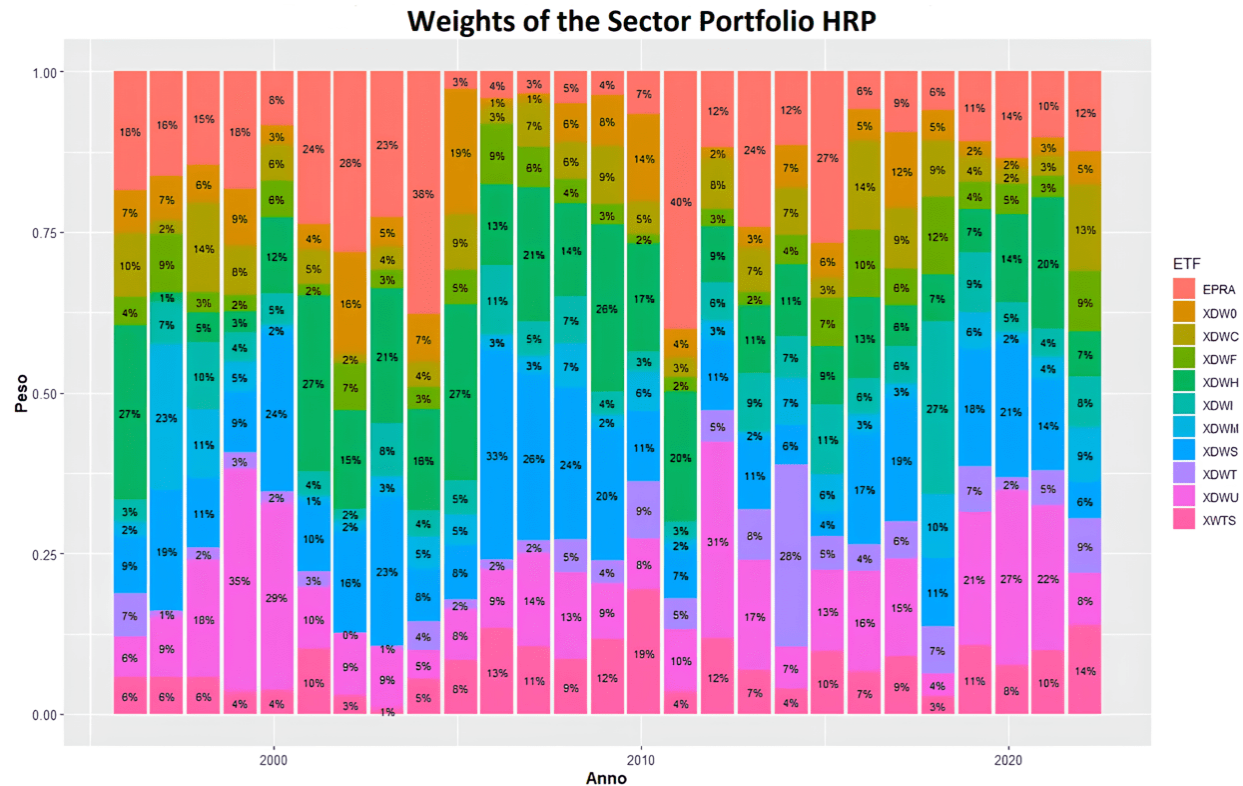
The lower section of the chart depicts the drawdowns of the two portfolios over the time period under consideration. The maximum drawdown for both happened during the subprime crisis in 2008-2009. The drawdown in the Sector Portfolio HRP was less severe, and the average drawdown was significantly lower (as seen in the previous table).

Let's look at the linear connection between the ETFs employed in the Sector Portfolio HRP throughout the time period under consideration:

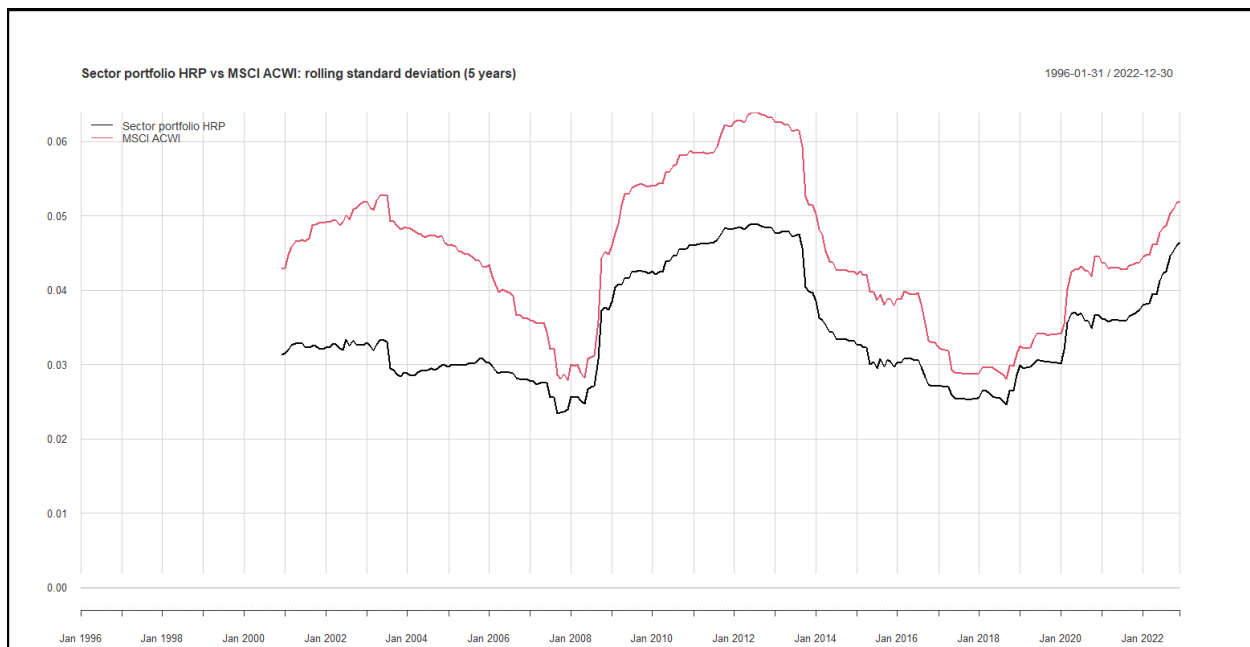


There are no major surprises: all the ETFs are positively correlated. EPRA is the ETF having the lowest correlation to the others. This discovery is also not surprising, given that the real estate sector has historically had the lowest correlation with the others, to the point that many authors consider it to be a distinct asset class.

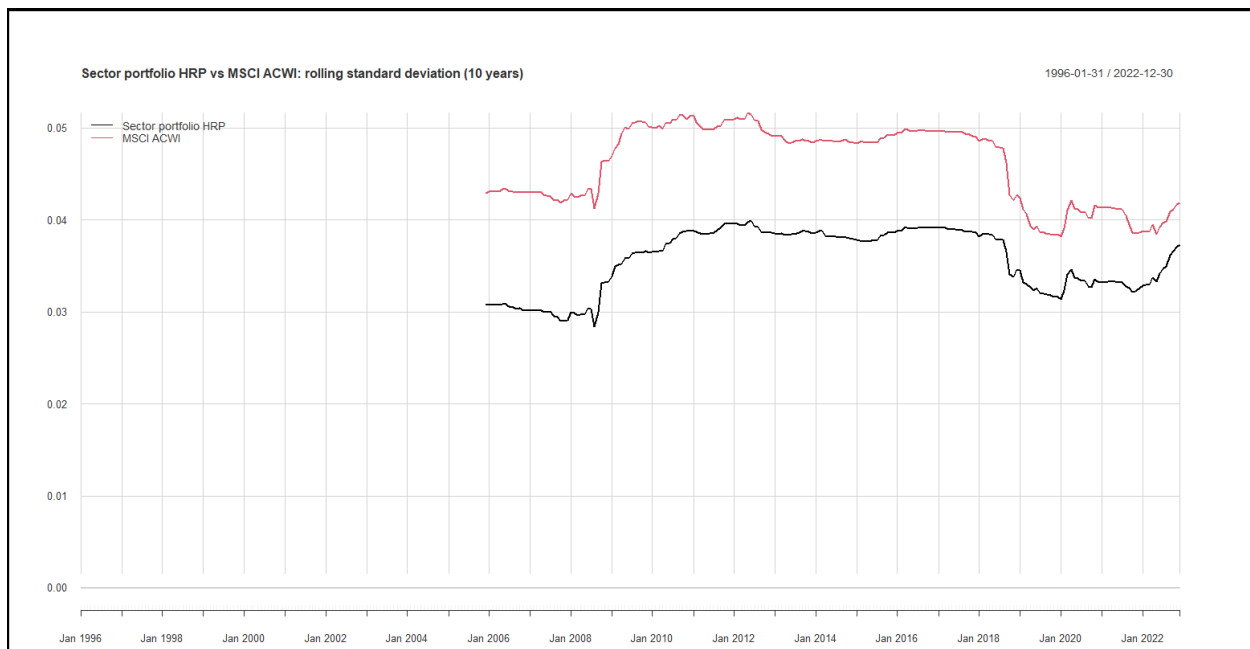
The chart below depicts the weight dynamics of the 11 ETFs that comprise the Sector Portfolio HRP between 1996 and 2022. Each vertical bar represents the ETF weights in a given year:



It might be interesting to compare the Sector Portfolio HRP versus the MSCI ACWI over shorter time intervals. Let us now examine the performance and volatility dynamics over rolling 5 and 10-year time periods. This method can reveal how the portfolio has responded to different economic cycles and provide a more complete picture of its performance and resilience in various market scenarios:

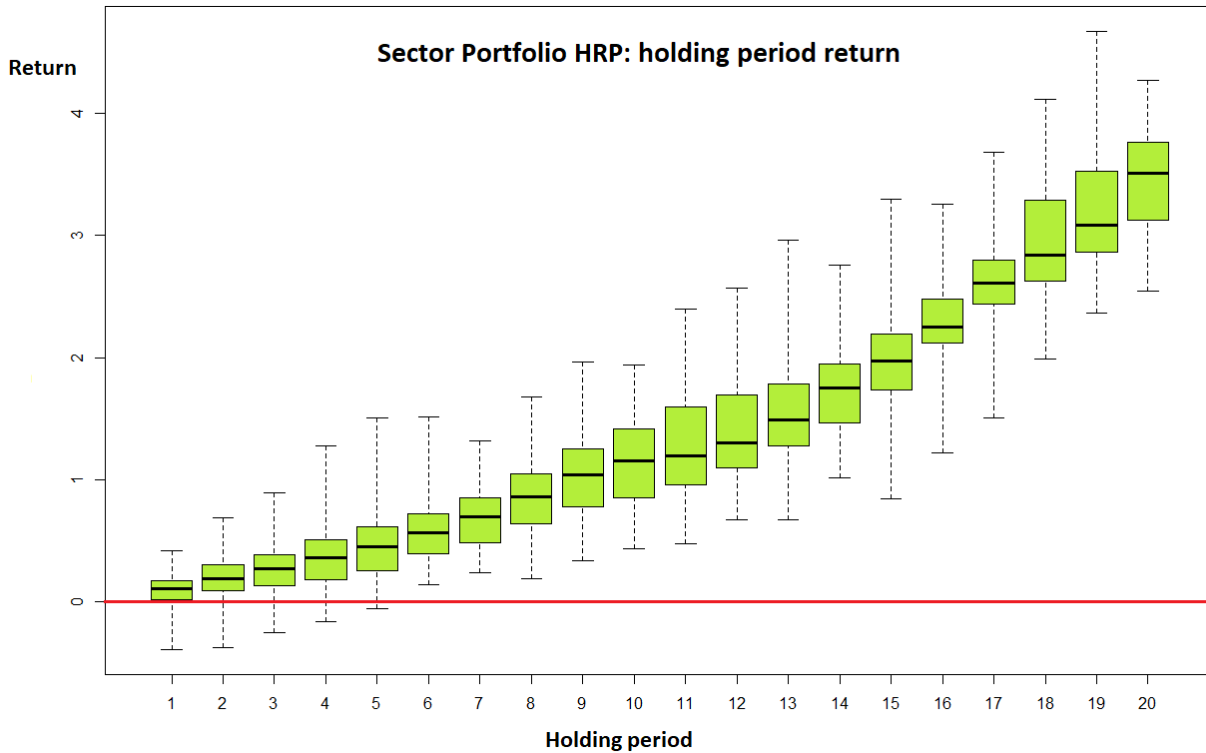


The rolling 10-year return and volatility are as follows:



The results are promising: the volatility of the Sector Portfolio HRP is consistently lower than the volatility of the MSCI ACWI, and its returns are higher in most of the sub-periods of 5 and 10 years.

We conclude our analysis with the presentation of the chart related to the cumulative holding period returns of the Sector Portfolio HRP:



Starting from the six-year holding period onward, the Sector Portfolio HRP has never recorded a negative return. Considering that this portfolio is entirely composed of stocks, this data appears quite reassuring.

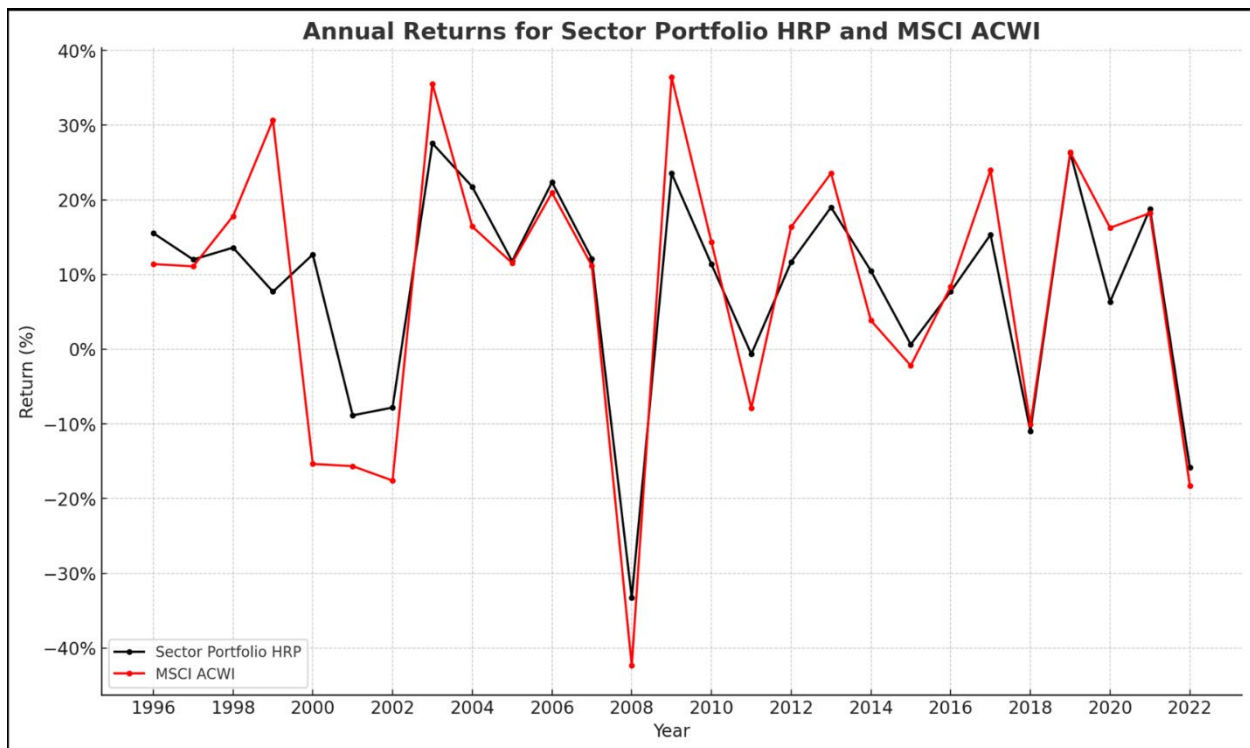
We end with a table that presents a year-by-year comparison of some performance metrics of the Sector Portfolio HRP and the MSCI ACWI. These metrics include return, volatility, and the Sharpe ratio:

Year	Performance	Sector Port. HRP	MSCI ACWI
1996	Return	15.54%	11.40%
	Std Dev	8.80%	8.07%
	Sharpe	1.7667	1.4131
1997	Return	12.00%	11.09%
	Std Dev	12.00%	13.88%
	Sharpe	1.0006	0.7994
1998	Return	13.60%	17.82%
	Std Dev	16.19%	20.79%
	Sharpe	0.8395	0.8570
1999	Return	7.70%	30.67%
	Std Dev	9.22%	12.44%
	Sharpe	0.8352	2.4647
2000	Return	12.68%	-15.36%
	Std Dev	7.71%	15.67%
	Sharpe	1.6442	-0.9800
2001	Return	-8.85%	-15.65%
	Std Dev	9.01%	19.44%
	Sharpe	-0.9826	-0.8049
2002	Return	-7.81%	-17.58%
	Std Dev	12.42%	18.72%
	Sharpe	-0.6288	-0.9393
2003	Return	27.56%	35.54%
	Std Dev	9.23%	12.20%
	Sharpe	2.9874	2.9142
2004	Return	21.74%	16.42%
	Std Dev	10.31%	8.80%
	Sharpe	2.1079	1.8671
2005	Return	11.78%	11.54%
	Std Dev	8.80%	8.94%
	Sharpe	1.3386	1.2918
2006	Return	22.36%	20.95%
	Std Dev	4.72%	8.34%
	Sharpe	4.7356	2.5123
2007	Return	12.12%	11.16%
	Std Dev	8.10%	9.97%
	Sharpe	1.4960	1.1197
2008	Return	-33.26%	-42.34%
	Std Dev	20.25%	25.58%
	Sharpe	-1.6422	-1.6555
2009	Return	23.54%	36.41%
	Std Dev	18.66%	24.12%
	Sharpe	1.2613	1.5099

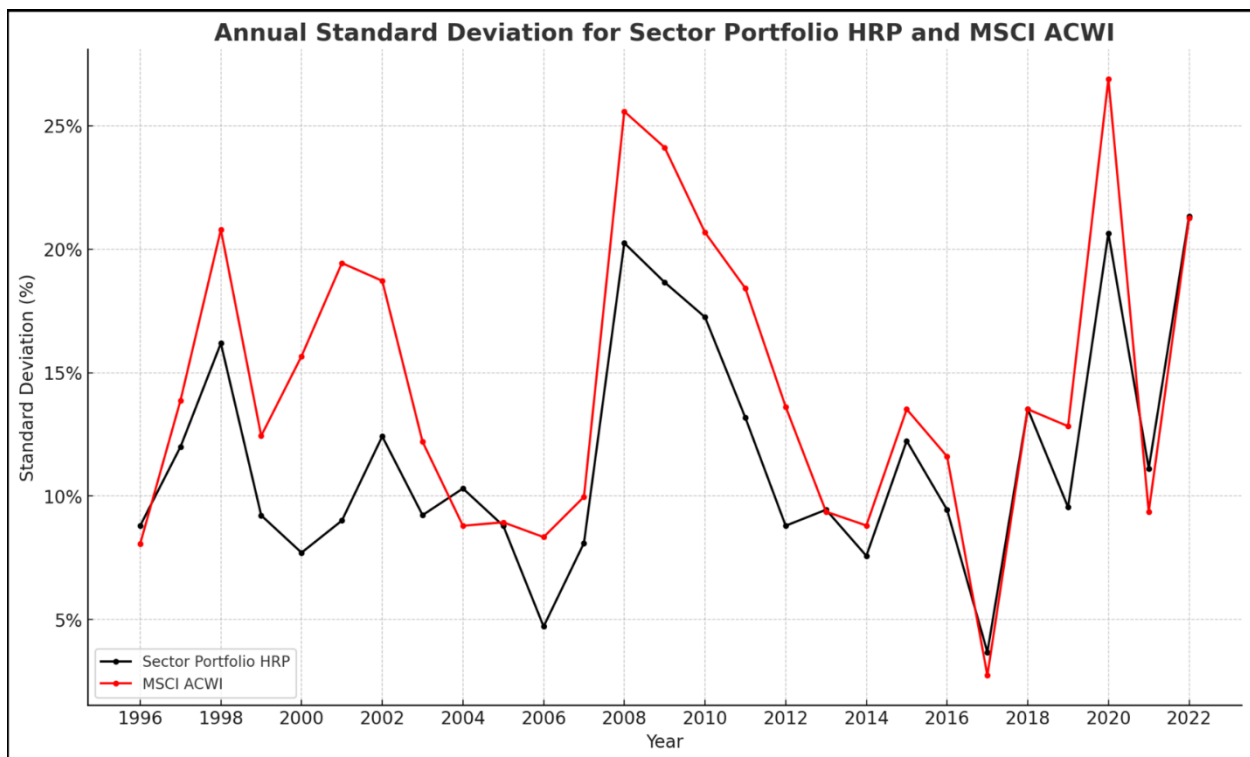
Year	Performance	Sector Port. HRP	MSCI ACWI
2010	Return	11.37%	14.35%
	Std Dev	17.25%	20.68%
	Sharpe	0.6594	0.6936
2011	Return	-0.65%	-7.89%
	Std Dev	13.18%	18.42%
	Sharpe	-0.0490	-0.4283
2012	Return	11.68%	16.38%
	Std Dev	8.80%	13.62%
	Sharpe	1.3268	1.2027
2013	Return	19.00%	23.55%
	Std Dev	9.45%	9.37%
	Sharpe	2.0112	2.5142
2014	Return	10.49%	3.84%
	Std Dev	7.58%	8.81%
	Sharpe	1.3826	0.4354
2015	Return	0.63%	-2.19%
	Std Dev	12.24%	13.52%
	Sharpe	0.0518	-0.1618
2016	Return	7.71%	8.36%
	Std Dev	9.45%	11.61%
	Sharpe	0.8157	0.7198
2017	Return	15.34%	23.95%
	Std Dev	3.68%	2.74%
	Sharpe	4.1698	8.7550
2018	Return	-10.95%	-10.08%
	Std Dev	13.53%	13.52%
	Sharpe	-0.8095	-0.7454
2019	Return	26.30%	26.35%
	Std Dev	9.55%	12.83%
	Sharpe	2.7528	2.0540
2020	Return	6.36%	16.25%
	Std Dev	20.65%	26.89%
	Sharpe	0.3079	0.6044
2021	Return	18.78%	18.22%
	Std Dev	11.12%	9.37%
	Sharpe	1.6891	1.9453
2022	Return	-15.84%	-18.27%
	Std Dev	21.34%	21.26%
	Sharpe	-0.7425	-0.8592

Negative Sharpe ratios are in grey because when returns are negative, they lose significance.

It may be useful to view the charts of the return and the standard deviation of the Sector Portfolio HRP and the MSCI ACWI from 1996 to 2022. This is the returns chart:



And this is the chart of the volatility:



The results of the Sector Portfolio HRP were decidedly above expectations: compared to the MSCI ACWI index, the higher return was accompanied by lower volatility and, overall, by lower risk (worst drawdown, average drawdown, average length, average recovery, Ulcer index, VaR, and CVaR).

Notably, the management fees of the employed ETFs are approximately 0.25% annually: this is undeniably modest in comparison to the fees imposed on actively managed funds, but nonetheless substantial given that the MSCI ACWI, being an index, is not subject to any expenses. In light of this consideration, the Sector Portfolio HRP's outcomes are even more remarkable.

In conclusion, the findings of our analysis emphasise the efficacy of the HRP model, thereby emphasising the criticality of conducting a more comprehensive examination of this approach within the domain of portfolio management. Its robustness indicates considerable potential. Implemented not only at the sectoral level but also on a geographic scale, the HRP model could represent a new frontier in asset allocation, which is currently preoccupied with the quest for miraculous factors and investment models that are becoming more complex and difficult to comprehend.

Conclusions

In this study, we examined and compared the strategy called “Sector Portfolio HRP” with the MSCI ACWI index, using a portfolio consisting of 11 sectoral ETFs.

This strategy, based on the annual rebalancing of the ETF weights employing the Hierarchical Risk Parity (HRP) methodology, achieved very promising results, showing a higher portfolio return compared to the MSCI ACWI index, accompanied by lower volatility.

It is essential to emphasise that the results obtained in our research are the product of specific market conditions and limited historical data. It is not possible to ensure that the Sector Portfolio HRP strategy will persistently produce comparable or even better outcomes in the future. As financial markets are susceptible to volatile dynamics and unforeseeable fluctuations, the performance of a given portfolio may be impacted in ways that are beyond our ability to precisely forecast.

Despite this, we believe that the Sector Portfolio HRP strategy represents an exciting avenue of research and a field of study deserving further exploration. The results achieved so far indicate significant potential, and further analyses could provide a deeper understanding of the underlying dynamics and conditions in which this strategy can be particularly advantageous.

In conclusion, alongside the excitement generated by the highlighted results, we recommend exercising caution when implementing this strategy in practice and advocate for additional investigation and evaluation to enhance comprehension of this approach.

Finance is a constantly evolving field, and ongoing research is essential to adapt to changing market conditions and develop robust and sustainable investment strategies in the long term.

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